

Autonomous Data Quality Investigation Agent

Assignee	Assigned By	Assigned On
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PROJECT DESCRIPTION

This project focuses on building an intelligent Agentic AI system that automatically detects, investigates, and resolves data-quality issues in business datasets—reducing the need for manual data investigation and improving data reliability.

The **Autonomous Data Quality Investigation Agent** acts as a centralized investigation system that uses specialized AI agents to analyze data quality, identify anomalies, investigate root causes, recommend corrective actions, and validate the results. By integrating data profiling, anomaly detection, schema analysis, and agent-based reasoning, the system provides an automated and explainable approach to maintaining high-quality data.

PROJECT SCENARIO

Organizations often depend on large datasets for dashboards, reporting, analytics, and business decisions. Missing values, duplicate records, inconsistent formats, schema changes, or abnormal data patterns can lead to incorrect reports and unreliable business insights.

For example, a company's sales dashboard suddenly reports a **20% drop in revenue**. Instead of requiring a data engineer to manually investigate the issue, the agentic system analyzes the dataset, checks for missing and duplicate records, examines schema changes and abnormal patterns, and investigates possible root causes. The system then recommends a correction, validates the corrected data, and generates an automated investigation report.

KEY FUNCTIONS & MODULAR COMPONENTS

Data Profiling Agent Automatically analyzes datasets to identify data types, missing values, duplicates, distributions, and overall data-quality characteristics.	Anomaly Detection Agent Identifies unusual patterns, abnormal values, sudden changes, and unexpected trends within the dataset.
Schema Analysis Agent Compares data structures and detects schema changes, inconsistent data types, missing columns, or unexpected fields.	Root Cause Investigation Agent Investigates detected data-quality issues by combining findings from multiple agents and determining the most likely source of the problem.
Correction Recommendation Agent Generates appropriate recommendations for resolving identified data-quality issues without directly modifying production data.	Validation Agent Validates the proposed correction by re-analyzing the affected data and checking whether the identified issue has been resolved.

Investigation Agent

Generates an explainable report containing detected issues, root-cause findings, recommended actions, validation results, and data-quality scores.

TOOLS & TECHNOLOGIES

Language

Python

Database

SQLite

Data Quality

Great Expectations

Agent Orchestration

LangGraph

AI / LLM

LLM API

Integration

MCP

Data Processing

Pandasc

API / Backend

FastAPI

Dashboard

HTML-CSS-Javascript